Real World Case Studies

1. The Smartphone Revolution

Descriptive — What happened?
Smartphone sales grew rapidly, transforming phones into everyday devices for communication, entertainment, and banking.

Diagnostic — Why did it happen?
Affordable devices, touchscreens, mobile internet, app stores, and improved cameras drove adoption.

Predictive — What might happen next?
AI-enabled smartphones and longer software support may shape future demand.

Prescriptive — What should we do?
Manufacturers should invest in useful AI features, battery efficiency, affordability, and durable devices.

2. COVID19 Pandemic

Descriptive:
Infections spread globally, overwhelming healthcare systems in many regions.

Diagnostic:
Respiratory transmission, limited initial immunity, and international travel accelerated the spread.

Predictive:
Infection trends and hospital admissions can help estimate future healthcare demand.

Prescriptive:
Improve disease surveillance, vaccination access, hospital capacity, and public-health communication.

3. RAM and Semiconductor Price Surge

Descriptive:
Memory shortages and rising prices affected the electronics supply chain.

Diagnostic:
Growing AI data-center demand, limited manufacturing capacity, and increased demand for high-bandwidth memory contributed to supply constraints.

Predictive:
Analysts can estimate future prices and shortages using production capacity, inventory, and demand forecasts.

Prescriptive:
Manufacturers should secure supply contracts, diversify suppliers, and optimize inventory.

4. EV Vehicle Adoption

Descriptive:
EV sales increased globally as electric mobility gained popularity.

Diagnostic:
Better batteries, government incentives, lower running costs, and improved charging infrastructure encouraged adoption.

Predictive:
EV demand and charging requirements may continue to grow.

Prescriptive:
Governments and businesses should expand charging networks, strengthen electricity infrastructure, and develop affordable EVs.

5. AI Data Center Expansion

Descriptive:
Data centers expanded to support cloud computing and AI workloads.

Diagnostic:
AI computing requirements, increased cloud adoption, and cooling needs drove infrastructure growth.

Predictive:
Electricity consumption and demand for computing hardware may increase significantly.

Prescriptive:
Operators should improve energy efficiency, plan power capacity, and optimize computing resources.